Jayraj Kiran Ladkat

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Education

B.Tech. in Computer Science and Engineering 10/2022 – 10/2026
MIT World Peace University, Pune, Maharashtra
Relevant Coursework: Data Structures & Algorithms, Distributed Systems, DBMS, Operating Systems, Computer Networks, OOP

Technical Skills

Languages:	Python, TypeScript, JavaScript, SQL, C/C++, Go (Basic)
Backend & Systems:	Node.js, FastAPI, Flask, PostgreSQL, Apache Kafka, NATS, Redis, REST APIs, WebSockets
AI & ML Systems:	Agentic RAG, LangGraph, LangChain, PyTorch, Mixed-Precision QAT, Pinecone
Cloud & DevOps:	Docker, AWS (ECS, S3), CI/CD Automation, Daytona Sandboxes, Linux, Git, Nginx, Prisma

Experience

TIGER-Lab, University of Waterloo *Waterloo, Canada (Remote)*
Software Engineering Intern (Research) 07/2026 – Present
- Engineered an automated multi-step generation pipeline (script → code → render → narration) for dynamic educational videos, reducing end-to-end animation synthesis latency by 45%.
- Architected an evolutionary prompt optimization engine to dynamically calibrate multi-agent LLM prompts, boosting pedagogical clarity and instructional alignment scores by 30%.

Publicis Sapient *Gurugram, Haryana, India*
Software Engineering Intern 08/2025 – 12/2025
- Optimized an AI-agent synthetic data-generation pipeline, slashing token consumption by 35% per run via prompt compression, structured caching, and schema enforcement.
- Enhanced an AI talent evaluation platform with client-side request prefetching and asynchronous assessment workers, cutting recruiter review turnaround time by 40%.

Elixir AI *Pune, Maharashtra, India*
Software Engineering Intern 06/2025 – 08/2025
- Developed full-stack features for the *Climate Smart Public Health Platform* (Harvard University & Malagasy Ministry of Health collaboration) using Flask, React, and PostgreSQL/PostGIS.
- Optimized spatial PostGIS queries across large-scale geospatial raster/vector datasets and containerized services using Docker, ensuring reproducible deployments presented to international stakeholders.

Centre for Development of Advanced Computing (C-DAC) *Pune, Maharashtra, India*
Cloud & Full-Stack Intern 01/2025 – 02/2025
- Integrated NATS pub/sub asynchronous messaging into microservice pipelines, eliminating synchronous HTTP blocking and decreasing peak-load request response times by 30%.

Selected Projects

Dexterio — Autonomous Educational Animation Generation Engine 2026
Python, TypeScript, Apache Kafka, Docker, Daytona, LangGraph, PostgreSQL [Code]
- Engineered an Agentic RAG engine orchestrating multi-step storyboard planning and modular Manim animation code generation with PostgreSQL-backed LangGraph checkpointers for fault-tolerant thread resumption.
- Architected a distributed asynchronous worker fleet using Apache Kafka and isolated Daytona/Docker VM sandboxes to securely compile, validate, and render untrusted generated Python code.

One Stop Deployment — Automated Microservice CI/CD Platform 2024
TypeScript, Docker, AWS (ECS, S3), Apache Kafka, Reverse Proxy, Node.js [Code]
- Developed a distributed one-click deployment engine automating build and containerization workflows for GitHub repositories using containerised Docker build workers and AWS ECS/S3.
- Implemented real-time build and deployment log streaming via Apache Kafka and designed a dynamic reverse-proxy for wildcard subdomain routing to deliver instant application preview URLs.

Demex — Distributed Full-Stack Search Engine 2024
TypeScript, Next.js, PostgreSQL, Node.js [Code]
- Built a high-throughput, robots.txt-compliant web crawler in TypeScript that recursively indexes web graph topologies into PostgreSQL using optimized relational graph schemas.
- Implemented a ranking pipeline combining TF-IDF lexical matching and PageRank link analysis; built a Next.js search UI supporting real-time query completion and sub-100ms result previews.

Selected Publications

- **Co-Adaptation over Allocation: KL-Guided Mixed-Precision Quantization-Aware Training** – *IEEE HiPC 2026* [Accepted Poster]
- **Real-World Scalability of PostgreSQL: Practical Techniques Use Case Study** – *IEEE ICBDS 2025* [IEEE Xplore 11377486]